[Experiment - 2]

[Determination of Current in Circuit using Mesh Analysis]

Aim :- To determine the current in circuit using Mesh analysis both theoretically and practically for a given DC circuit.

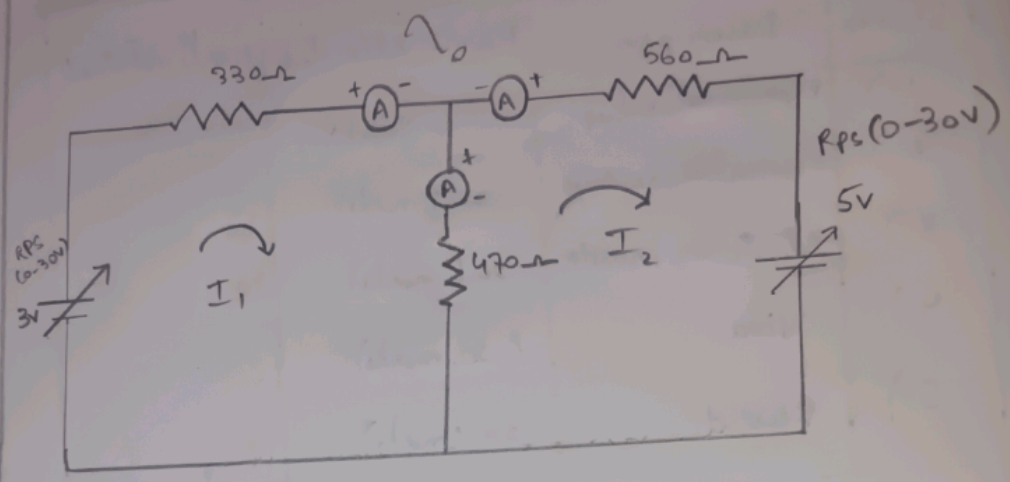
Apparatus Required :-

SL.NO	APPARATUS	SPECIFICATION	QTY
1.	Regulated Power Supply (RPS)	(0-30V)	1
2.	Ammeter	(0-10mA) MC	3
3.	Resistors	330Ω, 470Ω, 560Ω	Each 1
4.	Bread board	—	1
~~5.~~			

Theory :-

Multi-Source DC circuits may be analysed using a mesh current techniques. The process involves identifying minimum number of small loops such that every component exists in atleast one loop. KVL is then applied to each loop. The loop currents are referred to as mesh. Current interlocks or meshes with the surroundings loop currents. As a result, there will be a set of simultaneous equations created, and component voltages may be derived.

Circuit Diagram :



Sl.No	Description	Marks Allotted	Marks obtained
1.)	Experimental setup	20 (marks)	20
2.)	observation/readings	20 (marks)	20
3.)	calculations/results	20 (marks)	18
4.)	Viva	20 (marks)	18
5.)	Record	20 (marks)	18

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Calculations :-

loop 1 :-

$$330 I + 470 (I_1 - I_2) = 3$$

$$330 I_1 + 470 I_1 - 470 I_2 = 3$$

$$800 I_1 - 470 I_2 = 3 \rightarrow (1)$$

loop 2 :-

$$560 I_2 + 470 (I_1 + I_2) = 5$$

$$560 I_2 + 470 I_1 + 470 I_2 = 5$$

$$-470 I_1 + 1030 I_2 = -5 \rightarrow (2)$$

$$\begin{bmatrix} 800 & -470 \\ -470 & 1030 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 800 & -470 \\ -470 & 1030 \end{vmatrix} = 824000 - 220900$$

$$\Delta = 6,03,100$$

$$\Delta I_1 = \begin{vmatrix} 3 & -470 \\ -5 & 1030 \end{vmatrix} = 3090 - 2350$$

$$= 740$$

$$I_1 = \frac{\Delta I_1}{\Delta} = 1.2 A.$$

$$\Delta I_2 = \begin{vmatrix} 800 & 3 \\ -470 & -5 \end{vmatrix} = -4000 + 1410$$

$$= -2590$$

$$I_2 = \frac{\Delta I_2}{\Delta} = -4.3 A$$

$$[I_1 - I_2 = 1.22 + 4.3 = 5.52 mA]$$

Procedure :-

(i) Give Connections as per the circuit diagram.

(ii) Switch on the supply, vary the RPS and set a particular input voltage.

(iii) Note down the readings of ammeter and voltmeter and tabulate them.

(iv) Vary the RPS for different input and output voltages.

(v) Using the tabulated values, verify Kirchoff's laws practically and verify it Theoretically.

Result :-

Thus, the mesh analysis is verified practically and theoretically. The resultant current for voltage supply are, 3V / 5V.

a.) The current across 330 Ω resistor is 1.2 mA.

b.) The current across 560 Ω resistor is -4.3 mA.

c.) The current across 470 Ω resistor is 5.52 mA.